

Data Analytics Course Project N.06: Predicting Economy Fuel

Edoardo Silvio Gribaldo, Vojtěch Tichý

1 Introduction

The task of this project is to predict the cost of fuel based on the attributes of the car, its type, weight, size of its engine(s), its power etc. The chapters in this document are chronologically structured so that they follow the exact process of our progress on the project.

2 Exploratory analysis

The training dataset consists of 30 000 rows. The test dataset contains 7077 rows. Both datasets have 83 columns. The interpretation of the meaning of the columns was learned from the government website of the U.S. Department of Energy ¹.

Year

From the plot and the following table, we can see the data distribution of the training dataset by year. As shown in these plots, the test set was created by selecting mostly cars from 1985 to 1993 and some from the 2010s, while the training set contains cars from 1984 to 2017 with few cars from the years present in the test set. This is fundamental to note: the two sets will probably have different distributions. Mainly for this reason, we chose not to create a validation set as it would be incorrect to do validation on a set that has a distribution similar to the training set but different from the test set.

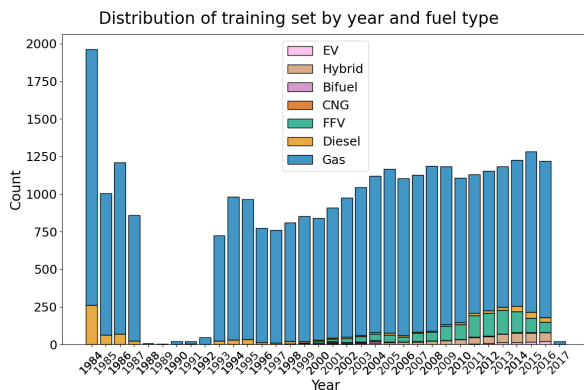


Figure 1: Training dataset distribution of cars by year and fuel type.

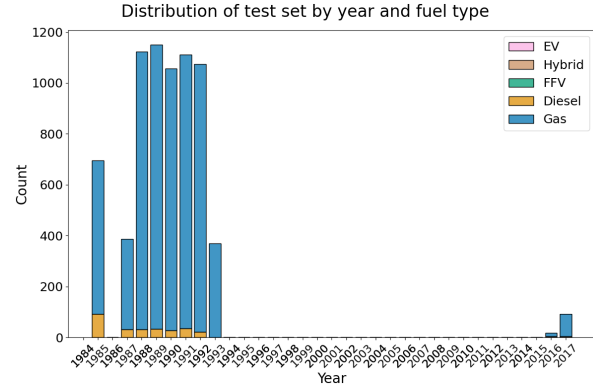


Figure 2: Test dataset distribution of cars by year and fuel type.

Car Make by Fuel Cost

We plotted with a bar chart the highest average fuel cost by car manufacturer. It is interesting to see that as expected, high performance cars have the highest average fuel cost. However, we decided not to take into consideration the column `make` because there are too many values and inside the same car manufacturer we could have completely different models of different sizes and performances that could confuse the algorithm. We also will not consider the `model` because similar models from different years may have much different fuel cost and also in this case the number of values is huge.

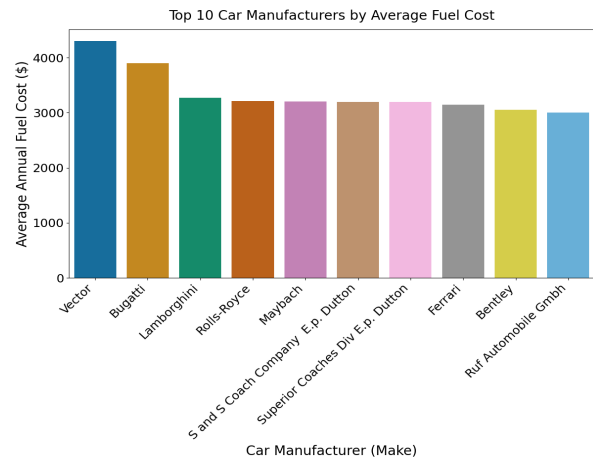


Figure 3: Top manufacturers ordered by the average fuel cost

¹<https://www.fueleconomy.gov/feg/ws/>

Start and Stop

We also explored the `startStop` column and saw that it had many `NaN` values. We decided to transform this column into a boolean column and fill the `NaN` values with 0. We assumed that most of the cars that had `NaN` values were built before the mainstream adoption of the start and stop control.

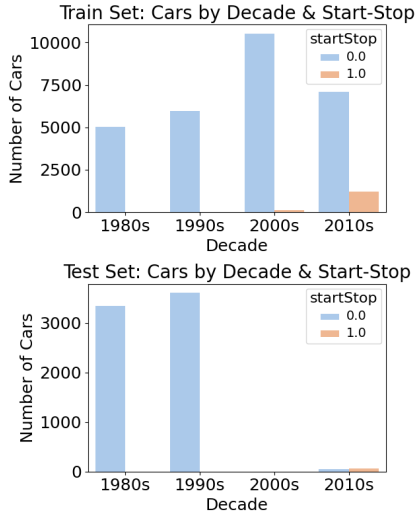


Figure 4: Training dataset distribution of cars by year and fuel type.

Cylinders

We also plotted the distribution of the number of cylinders to understand if there were any interesting patterns.

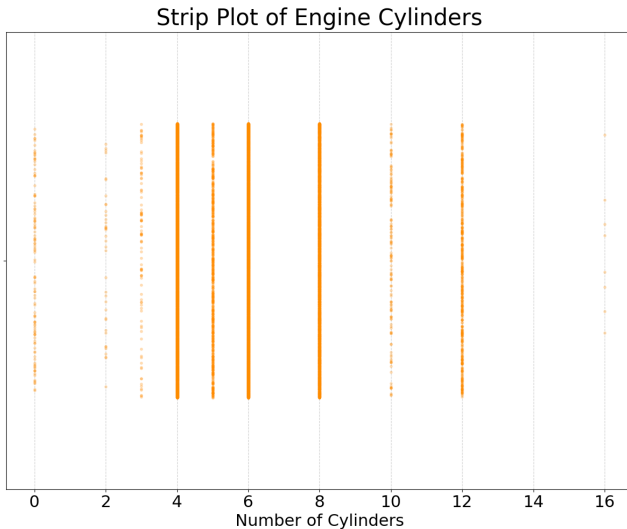


Figure 5: Plot of the distribution of cars by cylinders.

2.1 Outliers

Upon further analysis, we found some outliers in the `cylinders` column. In the whole dataset, there are 8 cars, which have 16 cylinders and all of those cars are Bugatti Veyron. Being this a single car model we decided to drop it because it is a really rare type of car it would only negatively influence the prediction model. There were also some few cars with 2 cylinders but since there were more we decided to keep them.

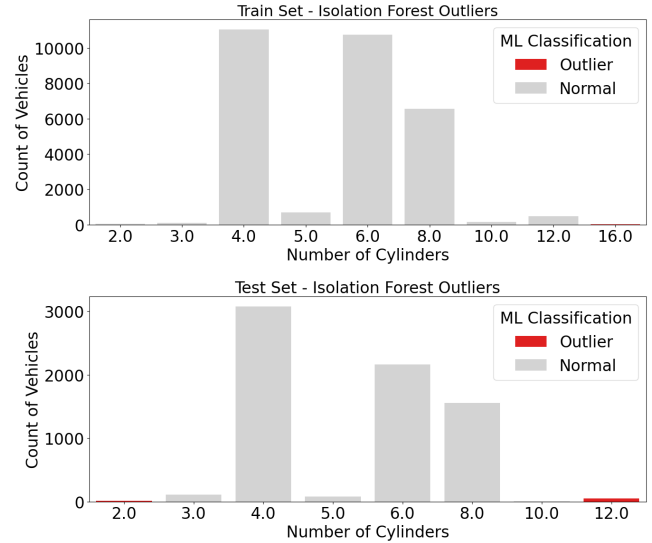


Figure 6: Histograms of distribution of cars based on the number of cylinders.

In the dataset, there were a lot of car manufacturers, we decided to drop all the rows related to car manufacturers which had 3 or less car models.

3 Data preprocessing

We had to pass to the preprocessing phase because a lot of columns needed preprocessing even before being analyzed. During this stage, the most important challenges were:

- dropping columns
- extracting information from columns
- condensing information by reducing the number of values in certain columns

3.1 Column drop

Out of the initial 83 columns we needed to select the relevant ones for our models. After reading the

²<https://www.fueleconomy.gov/feg/ws/>

meaning of the columns on the website ², we found that there were a lot of columns to be dropped, the most common reasons why we decided to drop one were:

- columns which had irrelevant and/or unnecessary information
- there was a high correlation with the target column, so keeping them would have been like "cheating"
- a significant number of values was missing and was not possible to impute the values
- most of the rows had just one value, so it was not bringing useful information

At first, we dropped columns which are completely irrelevant to our task, such as IDs, codes, timestamps of record creation and modification (in this case, most of the rows had just one timestamp). Next we dropped the columns describing chargers (for EV cars), engines, guzzler tax and transmissions since the number of values was limited and the number of unique values was too great for us to successfully interpret. We tried to keep our features minimal and clean as much as possible in order to make good inference.

Correlation analysis

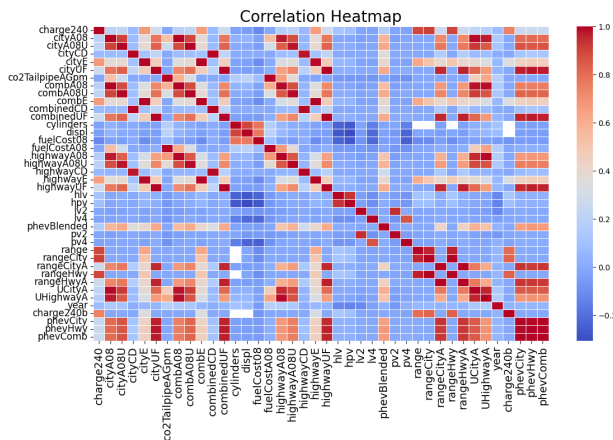


Figure 7: Correlation heatmap of training dataset.

Correlation analysis showed us a lot of candidate columns to be dropped. After further analysis, we decided to drop columns `cityA08` and `highwayA08` and keep `combA08` since it combines information of both columns. Moreover, we decided to drop all the columns ending with U (`city08U`, `highway08U`,

UCity, UHighway) that were repeated columns with unrounded ('U' at the end of column name) or unadjusted ('U' at the beginning) values. Eventually, we dropped the columns `co2`, (`co2TailpipeGpm`, `barrels08`, `feScore`, `ghgScore`, etc.), which also showed a high correlation with column `target`.

3.2 Column information extraction

From the columns left, we decided to extract some useful of information.

Vehicle size

Initially, we decided to compute the interior volume of each car and store the information in a separate column called `total_interior_volume`. This column combines the columns `hvp`, `pv2`, `pv4`, `hlv`, `lv2`, `lv4`. But then, we realized that this would not be sufficient since there are a lot of (almost 50%) missing values.

```
total_interior_volume
0      13693
110     575
105     485
100     481
103     467
111     462
```

Figure 8: total_interior_volume value counts.

Instead of using this metric, we utilized `VClass` attribute which has no missing value. However this column had a lot of unique values and some of the values classified the car based on its weight and others based on the car's size. Keeping in mind the information on the government's website, we decided to reduce the values to the values presented in the following pie charts:

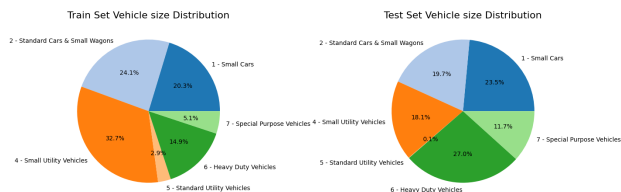


Figure 9: Distribution of VClass column after unique value reduction.

This way, we have 7 values representing the vehicle size based on its volume.

Transmission

Another column in need of revision was the transmission. Column `trany` contains a lot of information, but by default, all of the information is complicated to interpret. We decided to make 3 columns from this column: `automatic` (0 - manual, 1 - automatic), `numOfGears` (the number of gears that the transmission has) and `isCvt` (a boolean indicating whether the transmission is a CVT transmission or not.)

Now, there are 3 columns which make the interpretation of information more straightforward. The only issue was how to solve the CVT transmissions, since these are a special type of automatic transmission that does not have "normal" gears compared to other types of automatic transmissions. The CVT do not have gears by itself, it only simulates them. So, by default, we decided to fill all the CVT transmissions with 10 gears, even if some simulate another number of gears because it is just a feature to improve the user experience not a real mechanical difference.

Drive

For this column, it was also necessary to reduce the number of values since there was a great number of duplicate values. We first reduced them into: Front-Wheel Drive, Rear-Wheel Drive, All-Wheel Drive, Part time All-Wheel Drive. Here there was a few 2-Wheel drive cars which did not specified if they were rear-wheeled or front-wheeled. Because there were not many of them, we manually determined whether they were FWD or RWD. Finally, we transformed the information from this column into two boolean columns: `is_rear` and `is_front` (All-Wheel Drive cars have value 1 in both of these columns).

Power

For EV cars, we were able to determine the power of the car in kilowatts. This value can be found in `evMotor` column. We created a column `Power` which holds the total maximum power of the cars. If the car has 2 motors, the total power of the car is the sum of the powers of both engines. However, this was just an exercise of column processing because we did not create a model for electric vehicles since there were not enough EVs in the test set so it would have not been possible to test its accuracy.

fuelType

We can find 3 columns regarding fuel in the dataset: `fuelType`, `fuelType1`, `fuelType2` and one related column called `atvType`. Out of these columns, we create 2 new columns. The first column, `fuel`, contains a reduced number of unique values from the original `fuelType` column. The second column, `premium`, is a boolean column indicating whether the fuel was premium or regular. Since there was a number of missing or confusing values in these columns, after careful considerations we simplified the columns based on the following criteria:

- **Gas** has in the `fuelType` column either '*Regular*', '*Premium*' or '*Midgrade*' and it has *NaN* in the `evMotor` column
- **Diesel** has in the `fuelType` column value '*Diesel*'
- **FFV** has in the `atvType` column value '*FFV*'
- **CNG** has in the `fuelType` column value *CNG*
- **Bifuel** has in the `fuelType` column value '*Gasoline or natural gas*', '*Gasoline or propane*'
- **Hybrid** has in the `atvType` either '*Hybrid*' or '*Plug-in Hybrid*'
- **EV** has in the `atvType` column value '*EV*'

This distinction was made so that it reflects the differences in cars as accurately as possible. Moreover, each of these categories reflects a principle as how the car operates and every category is different from the others.

4 Models

For the models we decided to train models both on the whole dataset and on the separated subdatasets which were split based on the newly calculated column `fuel`. The models used in this project were: linear regression, random forests and support vector machines. We also experimented with an ensemble model for diesel and gas cars. With the Hill Climbing algorithm we determined the most useful features for model training and with the Shapley values we did some basic explainability.

For fuel-type specific models, we trained models on cars which have reasonable number of samples in the test set because otherwise we cannot test the

results. Only diesel and gas cars fulfilled the pre-requisites.

4.1 Feature selection

As mentioned earlier, we experimented with the Hill Climbing algorithm to try to select features which are important for the model while trying to identify those which worsen the performance of the model. The HC algorithm was tested and ran on Linear Regression model trained on the whole dataset.

The algorithm ran multiple times with different starting initializations. Every run ended with the same result and it showed that the `startStop` column should be dropped. After dropping it and retraining the model, we got a subtle improvement on the test set. The evidence of the `startStop` column not being useful is confirmed by the fact that the adoption of the start and stop control is relatively recent and the test set is mostly composed by old cars whose `NaN` value got filled with zero so it did not have great predictive power. As it is visible from the following plot, the most important features are if the fuel is premium or not, the value of the displacement and the number of cylinders of the car and then a little less important but still impacting are the number of gears if the car is rear-wheeled and the year of fabrication. This is inline with what we would expect.

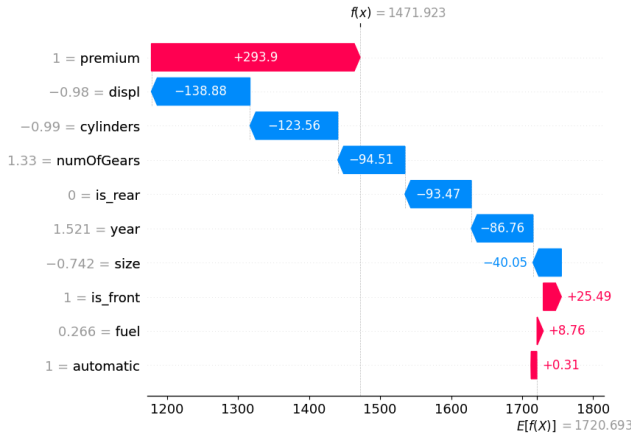


Figure 10: Explainability of attributes for Linear regression

4.2 Results

This table shows the results that we got with different models.

Segment	Model	R^2 Train	R^2 Test
Whole	LR	0.8022	0.7335
	SVM	0.7984	0.7242
	RF	0.9705	0.9026
Gas	LR	0.8634	0.6935
	SVM	0.8125	0.7505
	RF	0.9670	0.8865
Diesel	LR	0.8824	0.8401
	SVM	0.8603	0.8412
	RF	0.9771	0.8802
Ensemble	Ensemble	0.9329	0.8860

Table 1: Models performance

5 Summary

The main goal of this project was to predict fuel cost for various cars based on their specifications and features. At first, we performed exploratory analysis of the dataset. We filtered out all of the unnecessary columns, computed and extracted valuable information from the columns left. This allowed us to train and experiment with different models. We trained the models both on the whole dataset and on specific subdatasets grouped by fuel type; however, we did not train the models for all of the fuel types since some of them had too few rows in both training and test datasets, making it impossible to create a reasonably working model from the provided data. Next, we tried to combine the simpler models into an ensemble model, which performed surprisingly well. From the results, we can see that the Random forest model was the most accurate in all cases. On smaller datasets, the models (especially the LR) tended to overfit. The Linear regression's performance was very similar to the Support Vector Machine algorithm. Last but not least, we experimented with the Hill Climbing algorithm and Shapley values for Linear regression model to do feature engineering and understand feature importance. We reached satisfying results also considering that we had a test set which had a very different distribution with respect to the training set.